

3. Determine the number of integral solutions of $x_1 + x_2 + x_3 + x_4 + x_5 = 7$ where $x_i \geq 0, 1 \leq i \leq 4$.
4. Determine the number of integral solutions of $x_1 + x_2 + x_3 = 10$ where $x_i \geq 0, 1 \leq i \leq 3$.
5. Determine the number of integral solutions to the pair of equations $x_1 + x_2 + x_3 = 6, x_1 + x_2 + x_3 + x_4 + x_5 = 15, x_i \geq 0$
6. Determine the coefficient of xyz^2 in the expansion of $(x + y + z)^4$.
7. Determine the sum of all coefficients in the expansion of
 - a) $(w + x + y + z)^5$
 - b) $(2x - 3t + 5u + 6v - 11u + 3x + 2y)^{10}$
8. Determine the coefficient of $w^2x^2y^2z^2$ in the expansion of $(w + x + y + z + 1)^{10}$.
9. Determine the coefficient of $w^3x^2y^3z^2$ in the expansion of $(2w - x + 3y - 2z - 2)^{12}$.

Answers:

- ④ $4+8-1C_8 \quad 11C_8$
1. a) $11C_{12}$
 2. b) $13C_{11} + 11C_9 + 9C_7 + 7C_5$
 3. $11C_7$
 4. $12C_{10}$
 5. $9C_7$
 6. a) 4^5 b) 4^{10}
 7. $\frac{10!}{(2!)^5}$
 8. $\frac{12!}{3! \times 2! \times 3! \times 2! \times 2!}$
 9. $\frac{12! \times 3456}{(3!)^2 (2!)^3}$

2.6. Pigeonhole Principle

Statement

If m pigeons occupy n pigeonholes, $m > n$, then at least one pigeonhole will contain more than one pigeon.

Equivalently, if m objects are put in n boxes and $m > n$, then at least one box will contain two or more objects.

For example, an office employs 13 clerks, so that at least two of them have birthday during the same month. Here, we have 13 pigeons (office clerks) and 12 pigeon hole (month of the year).

Generalization of Pigeonhole principle

If m regions are accommodated in n pigeonholes and $m > n$, then one of the pigeonholes must contain atleast $\left\lfloor \frac{(m-1)}{n} \right\rfloor + 1$ pigeon, where $\left\lfloor \frac{(m-1)}{n} \right\rfloor$ denote the greatest integer less than or equal to $\frac{(m-1)}{n}$, which is a real number.

Worked Example. 2.6

Example 2.6.1

Show that if any four numbers from 1 to 6 are chosen, then at least 2 of them will add to 7.

Solution:

Here the pigeons constitute any four numbers from $\{1,2,3,4,5,6\}$ and the pigeon holes are the subsets $\{1,6\}$, $\{2,5\}$, $\{3,4\}$. Each of the four numbers chosen from 1 to 6 must belong to one of these sets. Since pigeons are greater than pigeon hole, by the Pigeonhole principle we can conclude that the two of the selected numbers belong the same set whose sum is 7.

Example 2.6.2

Given a group of 100 people, at minimum how many people were born in the same month?

Solution:

Here pigeons are people and pigeonholes are month of the year. So $m = 100$ and $n = 12$ then $\frac{(100-1)}{12} = 8$ By generalized pigeonhole principal people $8 + 1 = 9$ will born in the same month.

Example 2.6.3

Prove that if 6 integers are selected from the set $S = \{1,2,3,4,5,6,7,8,9,10\}$, then there are two integers such that one divides the other.

Solution:

For each $x \in S$, we may write $x = 2^k y$, with $k \geq 0$, $\gcd(2, y) = 1$. ie, $1 = 2^0 \cdot 1$, $2 = 2^1 \cdot 1$, $3 = 2^0 \cdot 3$, $4 = 2^2 \cdot 1$, $5 = 2^0 \cdot 5$, $10 = 2^1 \cdot 5$. Then y must be odd, so $y \in T = \{1,3,5,7,9\}$, where $|T| = 5$. Since 6 integers are selected from S , by the pigeon hole principle there are 2 distinct integers of the form $a = 2^m y$, $b = 2^n y$ for some $y \in T$ if $m < n$ then a divides b , otherwise if $n < m$ then b divides a .

Example 2.6.4

Let $S = \{3,7,11,15,19,\dots,95,99,103,107\}$. How many elements must we select from S to ensure that there will be at least 2 who sum is 110?

Solution:

Subdivide S in to 14 subsets whose sum is 110. 14 subsets are pigeon holes.

Let it be $\{3,107\}, \{7,103\}, \{11,99\}, \{15,95\}, \dots, \{51,59\}, \{55\}$.

By the pigeonhole principle, if we select at least 15 elements of S then we must have the elements in one of the 2-element subsets and their sum is 110. $\therefore$ the number of elements to be selected is 15.

Example 2.6.5

Prove that if any seven distinct numbers are selected from $\{1, 2, 3, \dots, 11\}$, from then the sum of two among them is 12.

Solution:

Here the pigeons consist of the seven numbers selected from the set $\{1, 2, 3, \dots, 11\}$. Pigeon holes are the subsets $\{1, 11\}, \{2, 10\}, \{3, 9\}, \{4, 8\}, \{5, 7\}, \{6\}$. Each of the seven numbers chosen from 1 to 11 must belong to one of these sets. By the Pigeonhole principle, we conclude that the two of the selected numbers belong to the same set whose sum is 12.

Example 2.6.6

Show that if 7 colors are used to paint 50 cars, at least eight cars will have the same color.

Solution:

Assume that 50 cars (pigeon) are assigned 7 colors (pigeonhole). Hence by the generalization of the Pigeonhole principle, at least $\left\lfloor \frac{(50-1)}{7} \right\rfloor + 1 = 8$ cars will have the same color.

Exercise 2.6

1. Prove that any subset of size 6 from the set $S = \{1, 2, 3, \dots, 9\}$ must contain 2 elements whose sum is 10.
2. Show that any 14 integers are selected from the set $S = \{1, 2, 3, \dots, 25\}$ there are at least 2 whose sum is 26.
3. Show that if any five numbers from 1 to 8 are chosen, then at least 2 of them will have a sum of 9.
4. Show that in a group of 13 children, there must be at least 2 children who were born in the same month.
5. Show that in any group of 8 people, at least 2 have birthdays which fall on the same day of the week in any given year.

Answers:

1. 5 – pigeon holes. Pigeons – 6.
2. Pigeons are the 14 integers and pigeon holes are the sets $\{1, 25\}, \{2, 24\}, \{3, 23\}, \dots, \{13, 12\}$. Selecting 14 pigeons from 13 pigeon holes, then at least 2 pigeons must be selected from one hole whose sum is 26.
3. 5 numbers are pigeons and $(1, 8), (2, 7), (3, 6), (4, 5)$ are pigeon holes.
4. Children are pigeons and month of the year is pigeon hole.
5. People are pigeons and days of the week are pigeon holes.